

Markscheme

Mathematics: analysis and approaches

Practice question bank

4. (a)

Given $a = 4 \times 10^{15}$ and $b = 9 \times 10^{12}$.

Calculate $C = \sqrt{a + b}$, giving your answer in the form $k \times 10^n$ correct to 3 significant figures.

match the powers of 10 before adding: $b = 9 \times 10^{12} = 0.009 \times 10^{15}$ (M1)

$$a + b \approx 4.009 \times 10^{15} \quad \text{A1}$$

rewrite with an even power of 10: $4.009 \times 10^{15} = 40.09 \times 10^{14}$ (M1)

attempt to take the square root (M1)

$$C = \sqrt{40.09 \times 10^{14}} = \sqrt{40.09} \times 10^7 \quad \text{A1}$$

$$C \approx 6.33 \times 10^7 \quad \text{A1}$$

Note: Do not accept a final answer given in calculator notation, e.g. $6.33E7$.

[6 marks]

(b)

Hence write C^2 in the form $k \times 10^n$, and verify it approximately equals $a + b$.

$$C^2 = (6.33 \times 10^7)^2 \approx 4.01 \times 10^{15} \quad \text{(M1)}$$

this matches $a + b$ from part (a), as required A1

[2 marks]